Technical Data Sheet

LOCTITE® LIOFOL LA 2798 / LA 7371

July 2019

Product description
LOCTITE LIOFOL LA 2798 / LA 7371 provides the following product characteristics:

Technology	Polyurethane
Product type	Laminating adhesive, aliphatic
Condition	Solvent-based
Components	Two components – requires mixing
Application	Film/Film and Film/Foil laminates
Performance	High
Cure	Room temperature cure after mixing
Appearance base LOCTITE® LIOFOL LA 2798	Transparent to yellow , clear to blear
Appearance hardener LOCTITE® LIOFOL LA 7371	Clear, colourless to yellow
Mix ratio, by weight (base:hardener)	6 : 1

Application areas
LOCTITE® LIOFOL LA 2798 / LA 7371 is suitable for film/film as well as film/foil laminates for retort applications. Transparent laminates produced with LOCTITE® LIOFOL LA 2798 / LA 7371 show practically no yellowing caused by UV-rays.

Product properties
Reaction of the two components produces an adhesive film with high polymer content which shows good adhesion on various substrates. The cured adhesive film is transparent, odorless, elastic and age-resistant.

Technical data

LOCTITE® LIOFOL LA 2798:

Solid Content, %	68 - 72
Viscosity, Brookfield - LVT, 20°C, mPa.s	
Spindle 3, Speed 12 rpm	2500 - 4500
Volumetric density, g/cm^3	1.1
Component	OH component
Solvent	Ethylacetate

LOCTITE® LIOFOL LA 7371:

Solid Content, %	68 - 72
Viscosity, Ford/DIN cup 4 mm, 20 °C, sec.	12 - 25
Volumetric density, g/cm^3	1.0
Component	NCO component
Solvent	Ethylacetate

Direction of use

Compliance with food packaging regulations
Under specific conditions LOCTITE® LIOFOL LA 2798 / LA 7371 is suitable for manufacturing of food packaging materials under the applicable FDA, EC-provisions and/or BfR recommendations.

A food contact statement with further details is available on request.

The responsibility for compliance with the specific conditions is with the converter and not with the adhesive supplier.

Preliminary statement
Prior to use it is necessary to read the **Safety Data Sheet** for information about precautionary measures and safety recommendations. Also, for chemical products exempt from compulsory labeling, the relevant precautions should always be observed. Please also refer to the local safety instructions and contact Henkel for analytical support.

Protective measures
As the adhesive contains polyisocyanates any contact with skin should be avoided. If skin gets in contact with LOCTITE® LIOFOL LA 2798 / LA 7371 please remove the adhesive immediately. Safety regulations with regard to handling of isocyanates have to be followed as per the guidelines issued by the International Isocyanates Institute. The maximum concentration of solvents in the air (TLV-value) has to be observed.

Dilution:
Suitable solvents are ethyl acetate, acetone and Methylethylketone provided that the water content does not exceed 300 ppm.
Do not use aromatic or alcoholic solvents.

Dilution table
In order to obtain a certain solids content, the following quantities of solvent have to be added.
The Ford/DIN cup (4 mm/20 °C) viscosities are based on an average viscosity of LOCTITE® LIOFOL LA 2798 / LA 7371.

LOCTITE® LIOFOL LA 2798, kg	LOCTITE® LIOFOL LA 7371, kg	Ethyl acetate, kg	Solid content, %	Viscosity, sec.
25	4.17	11.67	50	26- 30
25	4.17	16.20	45	18- 22
25	4.17	21.88	40	15- 19
25	4.17	29.17	35	13- 17
25	4.17	38.89	30	11 - 15

The recommended solid content is 30 to 40%.

Mixing
LOCTITE® LIOFOL LA 2798 has to be diluted under stirring with corresponding solvent to the desired application solids. Then the correct amount of hardener LOCTITE® LIOFOL LA 7371 has to be added under agitation.

Pot life
The ready-for-use adhesive (30 to 40%) in closed containers has a working time of about 6 hours without substantial viscosity increase. Overage adhesive mixtures as well as old adhesive in the glue pot affect the quality of the laminates adversely.

Application
LOCTITE® LIOFOL LA 2798 / LA 7371 can be applied with all dry laminating units with smooth or gravure rollers.

Application weight
The application weight may range from 3.5 to 4.5 g/m^2 dry depending on the film or foil combination and the intended use of the laminate. For printed laminates the weight has to be adjusted to higher weights accordingly, it should be approx. 1 to 2 g/m^2 higher than for unprinted films.

Working conditions
High laminating temperature together with a high line pressure of the calander improve the bond strength and optical appearance of the laminate.

Curing
Complete curing i. e. total resistance to severe environments will be achieved after 10 to 14days of storage at room temperature depending on film combination and intended use of the laminates. Best results before and after sterilisation are obtained if the laminates are stored at 40 to 50 °C for 7 to 10 days. Manufacture of triplex laminates is possible 24hours after duplex production. The laminates may be slit after 3 to 4 days. Processing of pouches should not start before the full curing.

Storage
To obtain the below mentioned shelf life in the original container we recommend for LOCTITE® LIOFOL LA 2798 a storage temperature of 20 to 30°C. LOCTITE® LIOFOL LA 2798 has to be protected from freezing.

Once opened, containers have to be tightly reclosed immediately and consumed within a short time span. When stored in a dry and cool place in the original unopened containers shelf life is as follows.

Shelf life	
LOCTITE® LIOFOL LA 2798, months	12
LOCTITE® LIOFOL LA 7371, months	9

Classification:
Please refer to the corresponding **Safety Data Sheet** for details on:
Hazardous Information
Transport Regulations
Safety Regulations

Film requirements
Film additives - particularly slip agent content in LDPE, MDPE and CPP films - packed goods, printing inks, pre-treatment and coating of films are of significant importance to the intended use of the laminates and may affect the performance properties either straight away or later.
Therefore, practical laminating tests as well as checks on the laminates are necessary before going into full scale production.
Further detailed information can be obtained from our "Technical Guide Line for Laminating with Liofol Laminating Adhesives".

Product range
Apart from LOCTITE® LIOFOL LA 2798 / LA 7371 we can offer the following products for the manufacture of film/foil laminates:

- Solventbased, solventless and waterbased laminating adhesives for flexible packaging and special applications
- Adhesives for high gloss lamination
- Primer for extrusion and PVDC-coating
- Coldseal adhesives
- Heatseal coatings
- Special cleaning agents

To obtain product specifications please ask for the relevant technical datasheets.

Additional information
Disclaimer
The information provided in this Technical data sheet (TDS) including the recommendations for use and application of the product are based on our knowledge and experience of the product as at the date of this TDS. The product can have a variety of different applications as well as differing application and working conditions in your environment that are beyond our control. We strongly recommend that you carry out your own prior trials to confirm such suitability of our product. Henkel is not liable for the suitability of our product for the production processes and conditions in respect of which you use them, as well as the intended applications and results. Any liability in respect of the information in the Technical data sheet or any other written or oral recommendation(s) regarding the concerned product is excluded, except if otherwise explicitly agreed and except in relation to death or personal injury caused by our negligence and any liability under any applicable mandatory product liability law.

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Reference 2

Henkel AG & Co. KGaA
40191 Düsseldorf, Germany
Phone: +49-211-797-0

Henkel Corporation
Cary, NC 27511, USA
Phone: +1-919-319-1933

Henkel Diadema
Diadema - SP, Brasil
Phone: +55-11-3205-8955

Henkel (China) Co.Ltd.
201203 Shanghai, China
Phone: +86-21-2891-8000

Henkel Jebel Ali FZCO
Dubai, UAE
Phone: +971-4-3893333